

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

The mobile technologies has fundamentally changed the way healthcare is delivered globally. Mobile health applications, particularly those focused on instant medical consultations, have become increasingly important in providing remote healthcare services [7]. As mobile devices become more ubiquitous, the development and effectiveness of mobile doctor applications have gained significant attention from researchers, healthcare professionals, and tech developers [8]. These applications, by offering immediate access to healthcare services, aim to address the barriers of traditional healthcare delivery systems such as waiting times, accessibility, and cost.

This chapter presents a comprehensive review of the existing literature on mobile doctor applications, telemedicine, and related technologies. It explores the theoretical underpinnings, empirical research, methodologies used, and key variables influencing the development and success of such applications. The goal is to identify critical factors that influence the design and effectiveness of mobile doctor apps, as well as to point out gaps in the literature that the current study aims to address.

2.2 Existing systems

2.2.1 Afyachap

Afyachap is a Tanzanian mobile app that connects users with doctors via in-app chat and offers health articles on nutrition and fitness. It has been lauded for facilitating easy consultations, but lacks structured features like appointment scheduling and dosage reminders. While it supports basic medical advice, it remains limited in comprehensive care management [9].

2.2.2 Afya Kiganjani (TanzMED)

Developed by Africa Healthtech LTD, TanzMED (also known as Afya Kiganjani) is an East African “health super app” that incorporates doctor chat, AI symptom analysis, pharmacy and lab bookings, home care services, pregnancy tracking, menstrual health tools, and medical content . Despite its rich features, TanzMED is heavily internet-dependent, lacks true offline functionality, and is relatively heavy in size (~90 MB) [10].

2.2.3 TanzMeds Pharmacy (Tanzmeds)

Tanzmeds is a local e-pharmacy service commonly affiliated with Tanzanian clinics. Though its primary function is medication ordering and delivery, it does not provide chat consultation or appointment booking features. Its strength lies in medicine accessibility but lacks engagement tools like symptom analysis or chat support.

2.2.4 Ada Health

Ada is an AI-driven symptom checker based on a question-and-answer diagnostic engine. It helps users self-assess health conditions but does not provide direct interaction with doctors, appointment systems, or medication reminders. The app requires continuous internet and functions only as an advisory tool [11].

2.2.5 Doctor on Demand

Doctor on Demand, a US-based telemedicine service, offers video and chat consultations with licensed doctors and their prescriptions. It also integrates behavioral health support. However, it requires strong internet, targets primarily mature markets, and lacks offline, medication-tracking, or local symptom tools [12].

2.3 Key Models

Most of the existing mHealth systems use one of the following models:

AI-Based Diagnosis Model – Used in apps like Ada Health, where users input symptoms and the system predicts possible illnesses.

Consultation-Based Model – Seen in Afyachap and Doctor on Demand, where users communicate directly with physicians.

Pharmaceutical Model – Implemented by Tanzmeds, where users primarily access and order medication.

Integrated Care Model – Rare in developing nations but ideal; it combines chat, appointments, diagnosis, and medication—what this project aims to achieve.

2.4 Gaps in Existing Knowledge

- i. Limited Offline Functionality: Most existing apps rely on active internet connections. In many parts of Tanzania, stable internet is still a challenge [13].
- ii. Lack of Integration: Most systems offer either consultation or medication services, but not both in a single app [14].
- iii. High Cost and Data Usage: Platforms like Doctor on Demand require high bandwidth and premium subscriptions, making them inaccessible to many users.
- iv. No Local Customization: Many global platforms are not customized for regional diseases, languages, or user behaviors [15].

The Mobile Doctor Instant Care Application is designed to address these limitations by offering an integrated system that functions offline, provides essential services like chat, symptom tracking, and reminders, and is tailored to the local healthcare context.